DevSecOps 與 AI 雙面刃的挑戰與機遇

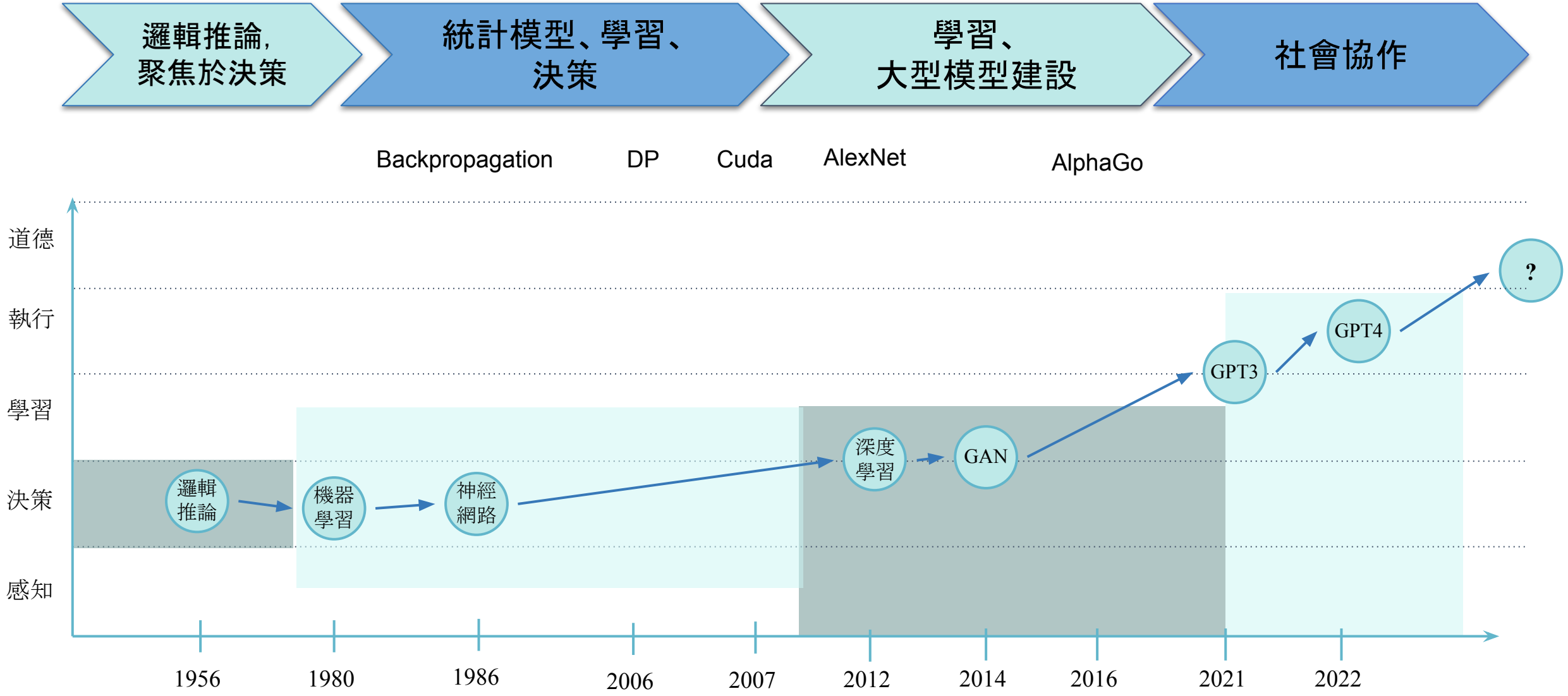
2024/07/09

Agenda

果核數位

- AI 發展概況
- 可視的未來近在眼前
- AI 系統的蓬勃發展可能帶來的隱憂

人工智慧的發展概況


邏輯推論，聚焦於決策
統計模型、學習、決策
學習、大型模型建設
社會協作
Backpropagation
DP
Cuda
AlexNet
AlphaGo
道德
執行
學習
決策
感知
邏輯推論
機器學習
神經網路
深度學習
GAN
GPT3
GPT4
?
1956
1980
1986
2006
2007
2012
2014
2016
2021
2022

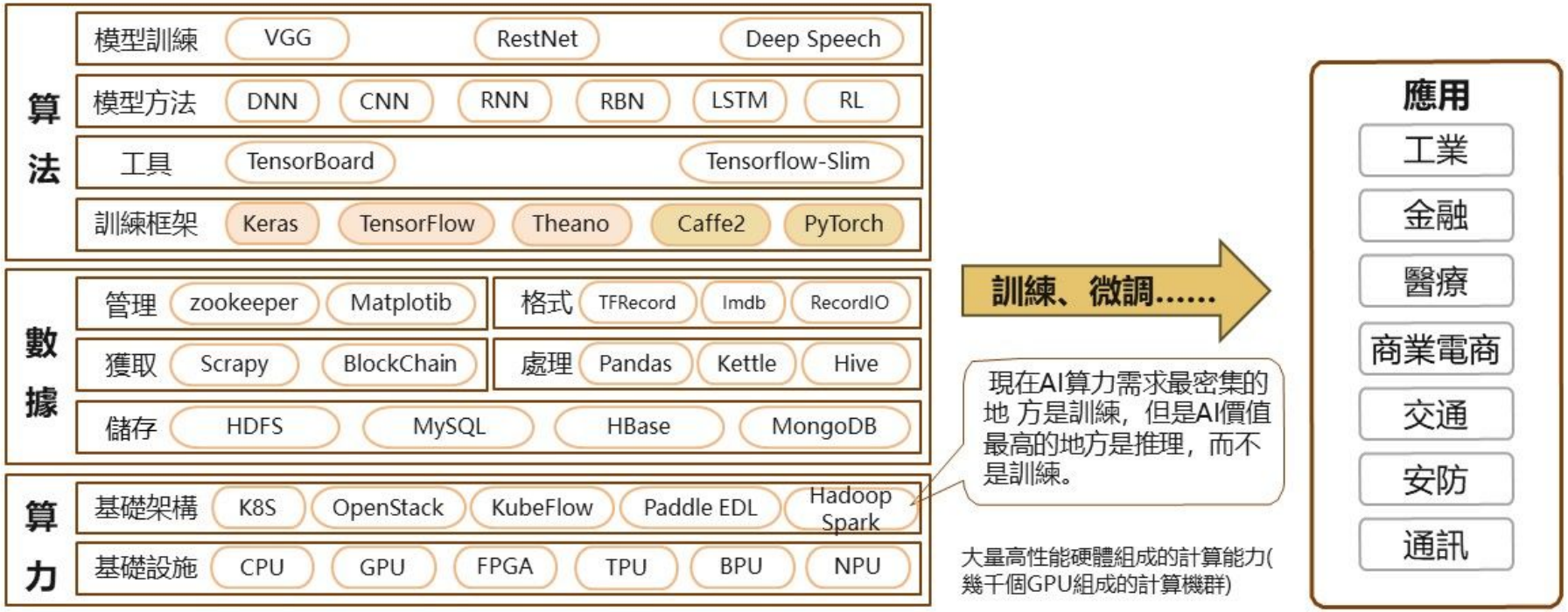
人工智慧的發展概況

果核數位

資料、算力、演算法作為人工智慧發展的核心三要素已經具備基礎條件

人工智慧三要素：數據（data）、演算法（algorithm）和算力（computing power）；

- 人工智慧(A)的快速發展依賴於三個核心要素:數據，演算法，算力。三個要素同時滿足了才能加速人工智慧的大發展。隨著人工智慧大模型規模變大以及普及應用，人工智能對能源的需求也在不斷加大，逐漸成為人工智能發展關鍵因素之一。



AI 核心要素

果核數位

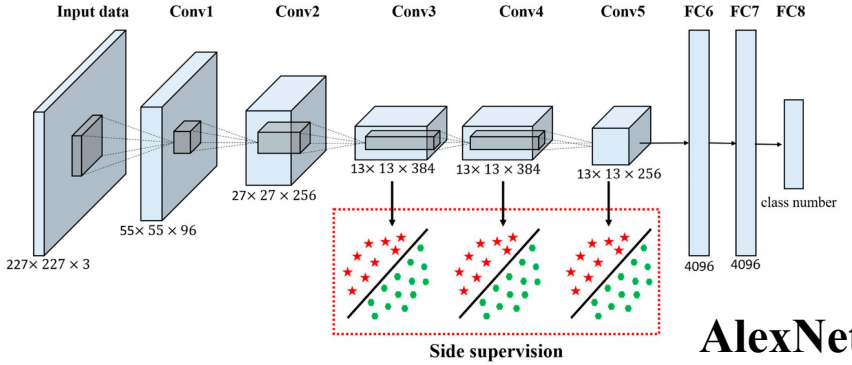
NVIDIA CUDA

TensorFlow

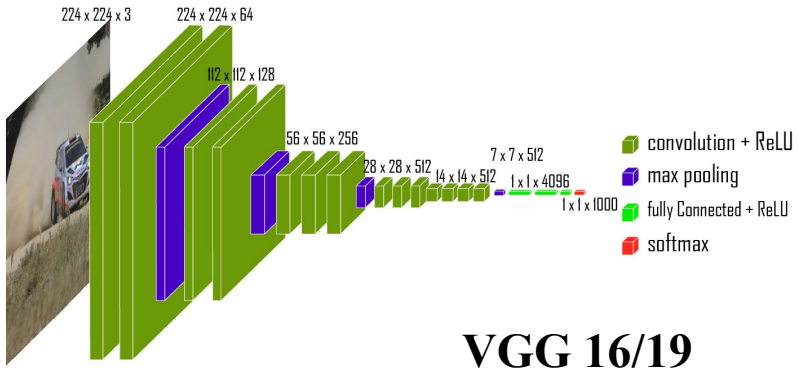
Caffe2

Keras

PyTorch



AlexNet



VGG 16/19

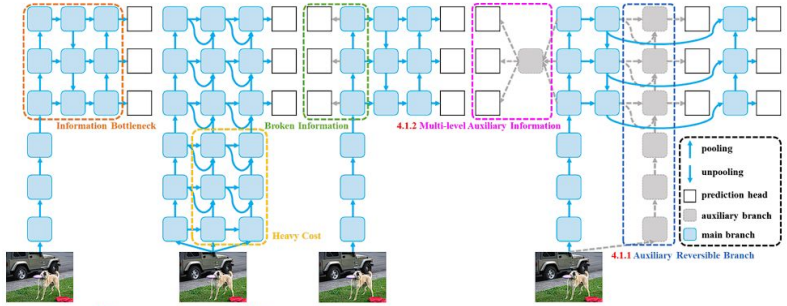
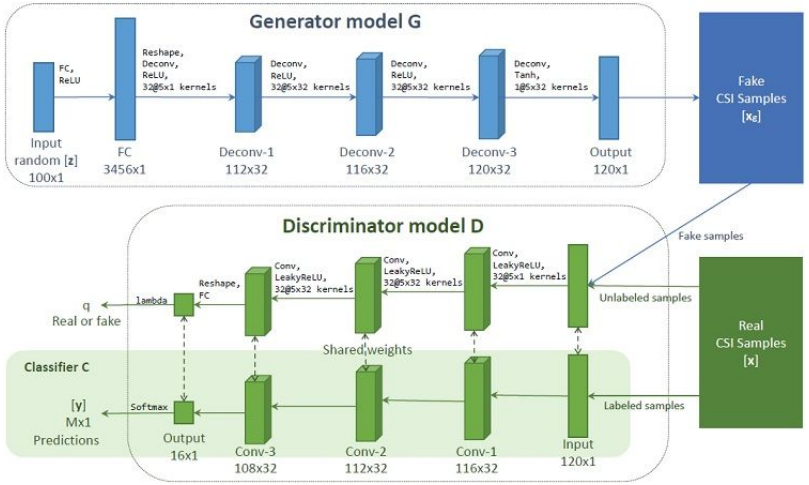
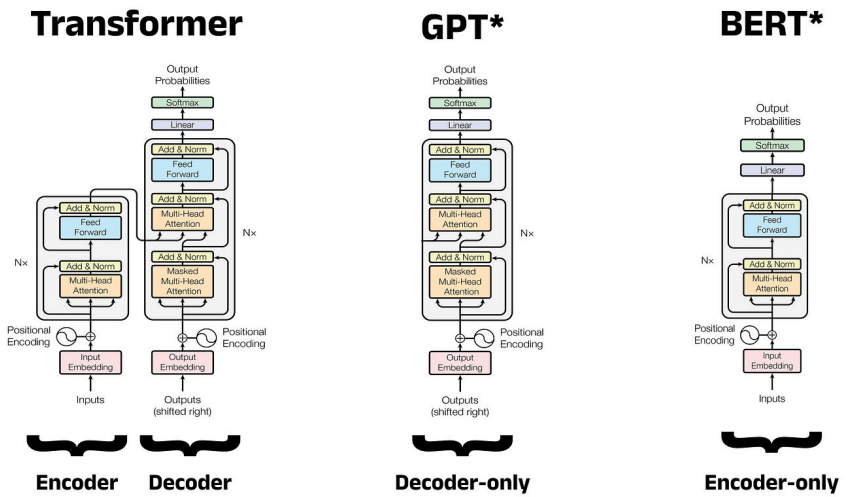


Figure 3. PGI and related network architectures and methods. (a) Path Aggregation Network (PAN)) [37], (b) Reversible Columns (RevCol) [3], (c) conventional deep supervision, and (d) our proposed Programmable Gradient Information (PGI). PGI is mainly composed of three components: (1) main branch: architecture used for inference, (2) auxiliary reversible branch: generate reliable gradients to supply main branch for backward transmission, and (3) multi-level auxiliary information: control main branch learning plannable multi-level of semantic information.

YOLO v9



Generative Adversarial Network



*Illustrative example, exact model architecture may vary slightly

AI應用常見情境

果核數位

- 藝術和設計
- 媒體和娛樂
- 自然語言處理
- 醫療和生物科學
- 商業和金融
- 工程和製造

大語言模型

- Encoder
- Decoder
- BERT
- GTP 3 ~ 4o

語意理解

- 資料分詞
- 詞性標註

影像識別

- 圖像分類
- 物件偵測

語音生成

- ASR
- TTS

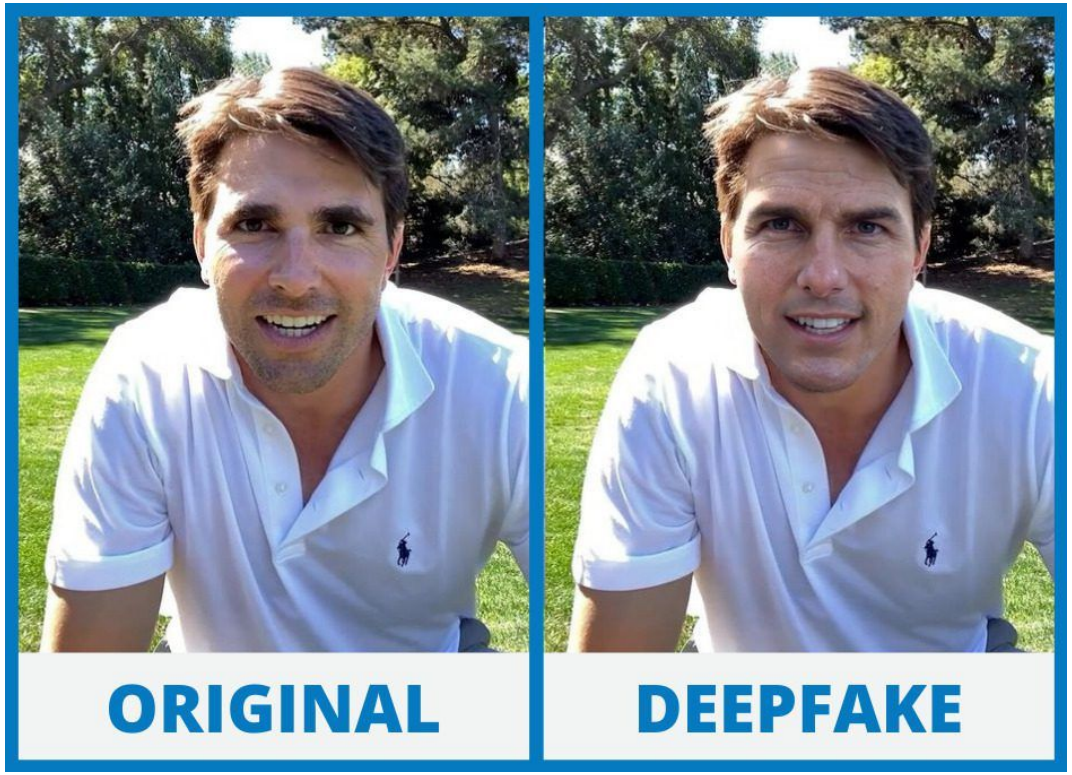
知識圖譜

- 既有文件資料
- RAG

機器學習

- 資料清理
- 特徵工程
- 模型營運管理
- 模型建構與驗證

AI 技術以假亂真



影片

AI語音3秒生成記者實測竟然親生媽媽都被騙！ @57ETFN

YouTube · 57東森財經新聞
1 個月前

疑AI深偽仿聲詐騙專家：講話5秒即可擬真7成｜20231212 公視 ...

YouTube · 公視新聞網
2023年12月12日

AI假兒子語音太逼真來電騙走老夫妻2萬美元

YouTube · 台視新聞 TTV NEWS
2023年3月16日

詐團最新武器！「AI 語音詐騙」攻陷全球，我們該怎麼辦？｜志 ...

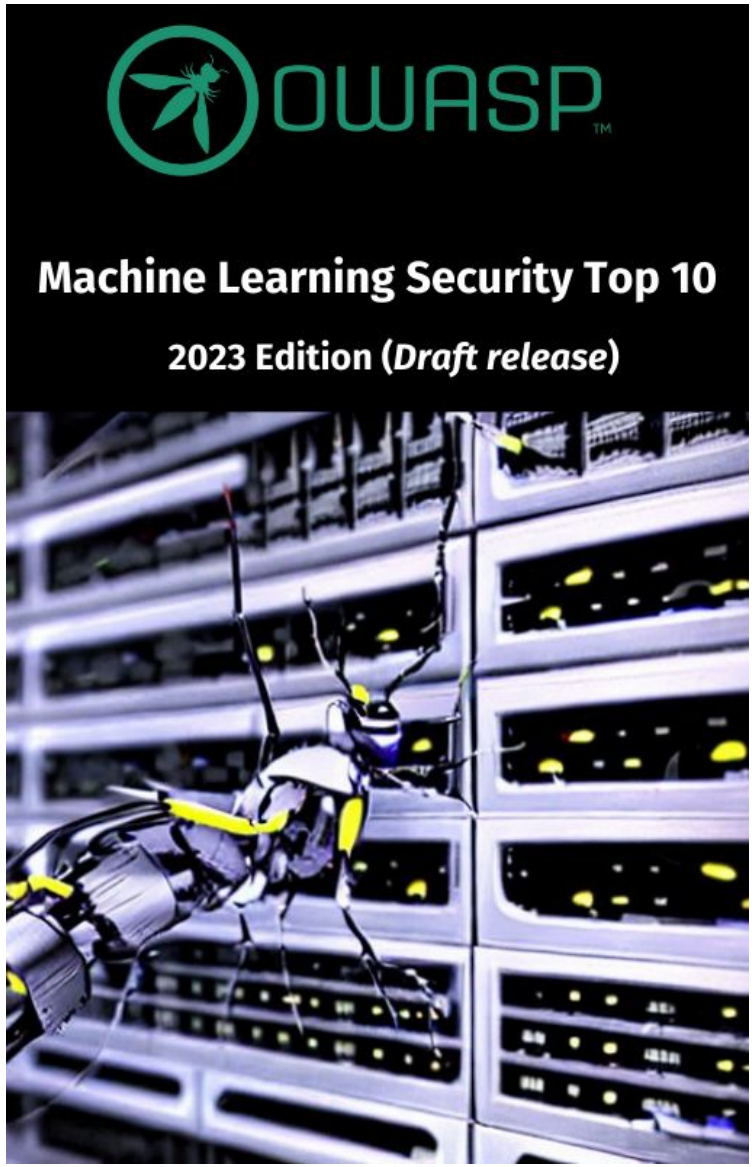
YouTube · 志祺七七 X 圖文不符
2023年9月3日

全部顯示 →

ref : https://vocus.cc/article/649ae36dfd8978000103 94f8

OWASP ML Top 10

- ML01:2023 Input Manipulation Attack
- ML02:2023 Data Poisoning Attack
- ML03:2023 Model Inversion Attack
- ML04:2023 Membership Inference Attack
- ML05:2023 Model Theft
- ML06:2023 AI Supply Chain Attacks
- ML07:2023 Transfer Learning Attack
- ML08:2023 Model Skewing
- ML09:2023 Output Integrity Attack
- ML10:2023 Model Poisoning



OWASP LLM Top 10

果核數位

LLM01 Prompt Injection

LLM01: Prompt Injection

Prompt Injection Vulnerability occurs when an attacker manipulates a large...

Read More

April 10, 2024

LLM02 Insecure Output Handling

LLM02: Insecure Output Handling

Insecure Output Handling refers specifically to insufficient validation, sanitization, and...

Read More

April 10, 2024

LLM03 Training Data Poisoning

LLM03: Training Data Poisoning

The starting point of any machine learning approach is training...

Read More

April 10, 2024

LLM04 Model Denial of Service

LLM04: Model Denial of Service

An attacker interacts with an LLM in a method that...

Read More

April 10, 2024

LLM05 Supply Chain Vulnerabilities

LLM05: Supply Chain Vulnerabilities

The supply chain in LLMs can be vulnerable, impacting the...

Read More

April 10, 2024

LLM06 Sensitive Information Disclosure

LLM06: Sensitive Information Disclosure

LLM applications have the potential to reveal sensitive information, proprietary...

Read More

April 10, 2024

LLM07 Insecure Plugin Design

LLM07: Insecure Plugin Design

LLM plugins are extensions that, when enabled, are called automatically...

Read More

April 10, 2024

LLM08 Excessive Agency

LLM08: Excessive Agency

An LLM-based system is often granted a degree of agency...

Read More

April 10, 2024

LLM09 Overreliance

LLM09: Overreliance

Overreliance can occur when an LLM produces erroneous information and...

Read More

April 10, 2024

LLM10 Model Theft

LLM10: Model Theft

This entry refers to the unauthorized access and exfiltration of...

Read More

April 10, 2024

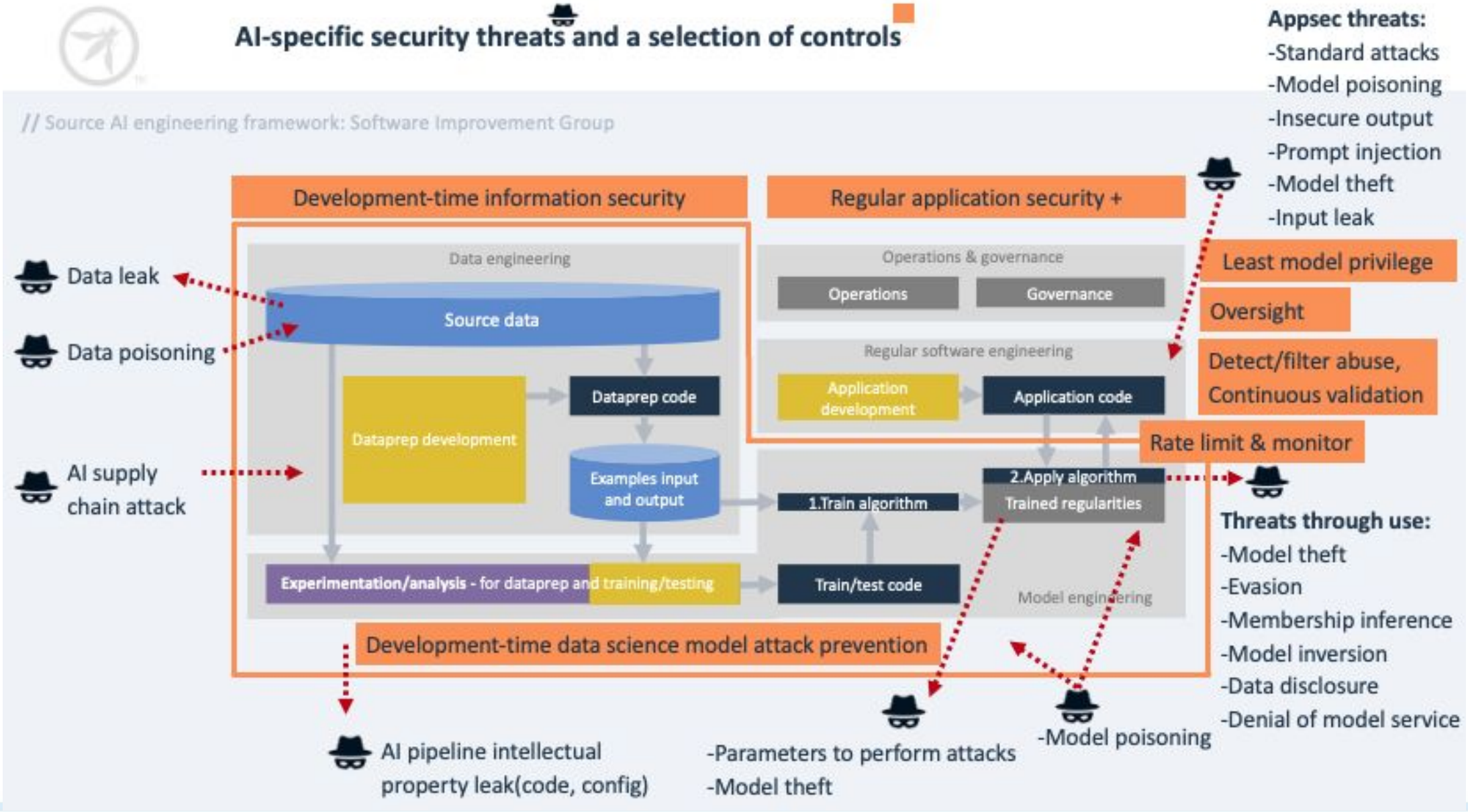
現在小孩不應該學習寫程式如何使用 AI 更為重要

在過去 10 至 15 年間，絕大多數在這論壇上的人，都曾堅信該讓年輕人學習電腦科學、掌握編寫電腦程式的技能是非常重要。然而，事實上，情況幾乎完全相反。

我們的責任是發展出一種計算技術，讓大家無需撰寫程式碼，透過人類語言就能直接編程語言。如今，**全世界的每一個人都能成為程式設計師**，這正是人工智慧帶來的奇蹟。

https://today.line.me/tw/v2/article/GgpYELY

AI 模型威脅模型



https://github.com/OWASP/www-project-ai-security-and-privacy-guide/blob/main/owaspaiexchange.md

實際案例

預計放 coploit 的圖片

實際案例

果核數位

Hello.java

create Hello class with sql connection build a login check sql query and user account and password from termail arguments

GitHub Copilot

已進行變更。

接受 捨棄

接受或捨棄 1 個變更

```java
import java.sql.*;

public class Hello {
    Run | Debug
    public static void main(String[] args) {
        // Check if the correct number of arguments is provided
        if (args.length != 2) {
            System.out.println(x:"Please provide the user account and password as command line arguments.");
            return;
        }

        String userAccount = args[0];
        String password = args[1];

        // JDBC connection variables
        String url = "jdbc:mysql://localhost:3306/mydatabase";
        String username = "root";
        String dbPassword = "password";

        try {
            // Establish the connection
            Connection connection = DriverManager.getConnection(url, username, dbPassword);

            // Prepare the SQL statement
            String sql = "SELECT * FROM users WHERE account = ? AND password = ?";
            PreparedStatement statement = connection.prepareStatement(sql);
            statement.setString(parameterIndex:1, userAccount);
            statement.setString(parameterIndex:2, password);

            // Execute the query
            ResultSet resultSet = statement.executeQuery();

            // Check if the login is successful
            if (resultSet.next()) {
                System.out.println(x:"Login successful!");
            } else {
                System.out.println(x:"Invalid credentials. Please try again.");
            }

            // Close the resources
```

Copilot

實際案例

果核數位

```java
public class Hello {
    Run | Debug
    public static void main(String[] args) {
        // Check if the correct number of arguments is provided
        if (args.length != 2) {
            System.out.println(x:"Please provide the user account and password as command line arguments.");
            return;
        }

        String userAccount = args[0];
        String password = args[1];

        // JDBC connection variables
        String url = "jdbc:mysql://localhost:3306/mydatabase";
        String username = "root";
        String dbPassword = "password";

        try {
            // Establish the connection
            Connection connection = DriverManager.getConnection(url, username, dbPassword);

            // Prepare the SQL statement
            String sql = "SELECT * FROM users WHERE account = " + username + " AND password = " + dbPassword;



            connection.close();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
```

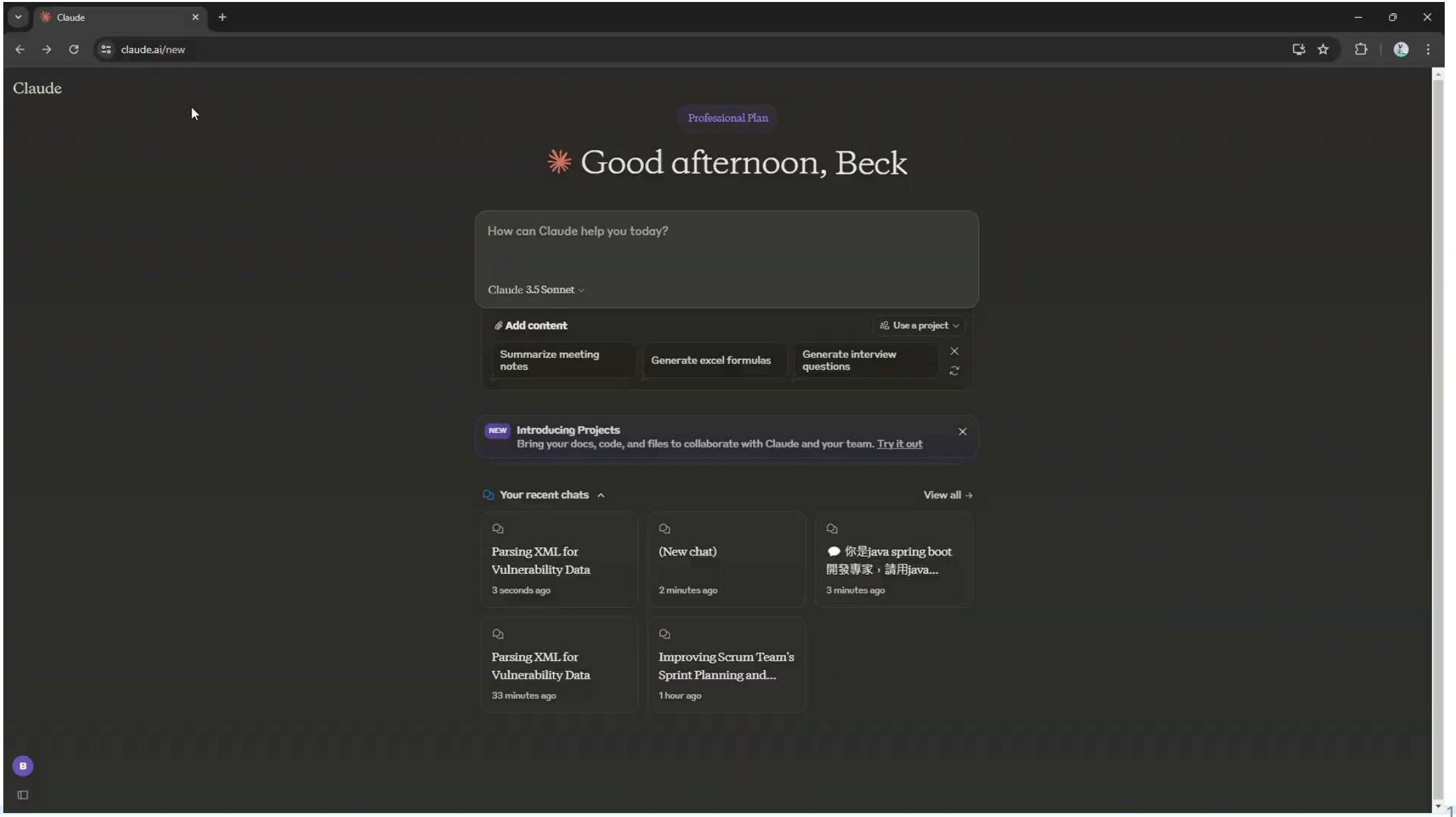
Copilot

實際案例

果核數位

預計放 amazon Q Developer 的圖片

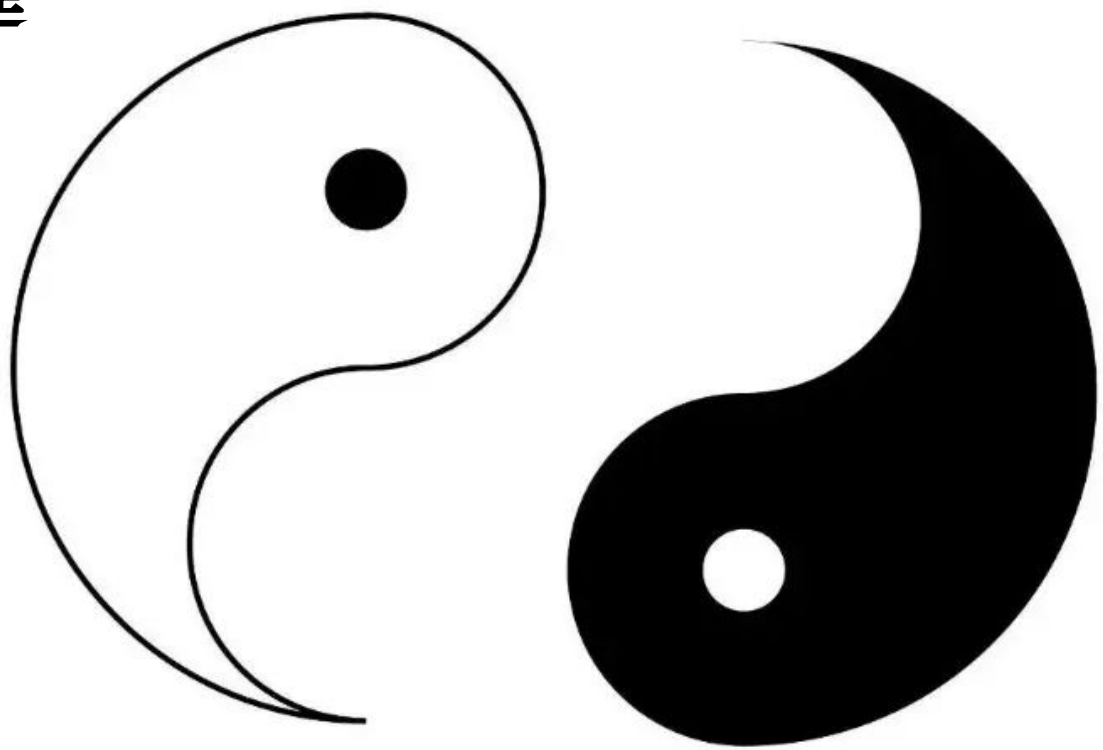
實際案例


Claude
claude.ai/new
Claude
Professional Plan
Good afternoon, Beck
How can Claude help you today?
Claude 3.5 Sonnet
Add content
Use a project
Summarize meeting notes
Generate excel formulas
Generate interview questions
NEW
Introducing Projects
Bring your docs, code, and files to collaborate with Claude and your team. Try it out
Your recent chats
View all
Parsing XML for Vulnerability Data
3 seconds ago
(New chat)
2 minutes ago
你是java spring boot 開發專家，請用java...
3 minutes ago
Parsing XML for Vulnerability Data
33 minutes ago
Improving Scrum Team's Sprint Planning and...
1 hour ago

挑戰與機會

果核數位

AI 模型能力演進



安全開發流程
與
安全治理的重要性